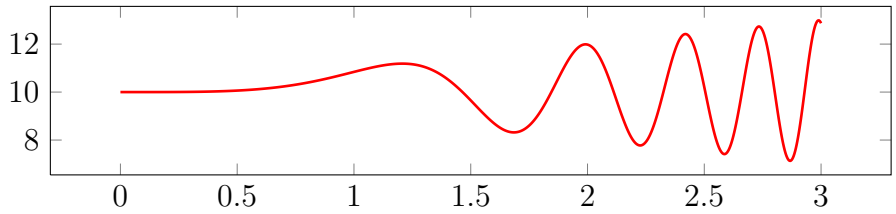
CISS240: Introduction to Programming
Quiz q1006

Name: eagraham1@cougars.ccis.edu Score:

This is a closed-book, no compiler, 5 minute quiz.

Q1. The following is the graph of $y = x^2 \sin(x^3) + 10$:



Using exactly 1000 rectangles, compute the area under the above curve from $x = 0.5$ to $x = 2.5$. The area of each rectangle must be computed using the method in the notes (i.e., use the left endpoint of the rectangle to determine the area of the rectangle). The printout of the area must be in fixed point format with 5 decimal places. Skeleton code is provided.

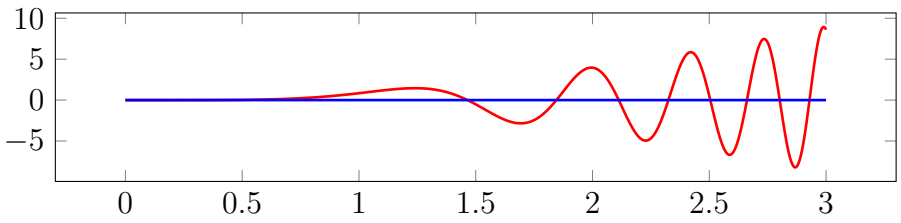
Answer:

```
#include <iostream>
#include <iomanip>
#include <cmath>

int main()
{
    double area = 0.0;                // the required area
    double a = 0.5, b = 2.5;          // bounds
    int n = 1000;                     // number of rectangles
    double dx = (b - a) / n;          // width of each rectangle

    for (int i = 0; i < n; ++i)
    {
        double x = a + i * dx;
        double y = x * x * sin(x * x * x) + 10;
        area += y * dx;
    }
    std::cout << std::fixed << std::setprecision(5) << area << std::endl;
    return 0;
}
```

Q2. The following is the graph of $y = x^2 \sin(x^3)$:



Notice that there's a root at approximately $x = 1.7$. Use exactly 1000 equally spaced points from $a = 1.6$ to $b = 2.0$ to compute an approximation to this root. The printout of the value x of the root must be in fixed point format with 5 decimal places. Skeleton code is provided.

Answer:

```
#include <iostream>
#include <iomanip>
#include <cmath>

int main()
{
    double area = 0.0;                  // the required area
    double min_y;                       // the running minimum y for given x
    double a = 0.5, b = 2.5;            // bounds
    int n = 1000;                       // number of points
    double dx = (b - a) / n;            // width of each rectangle

     for (int i = 0; i <= n; ++i)
    {
        double x = a + i * dx;
        double y = x * x * sin(x * x * x);

        if ((y > 0 && y < min_y) || (y < 0 && -y < min_y))
        {
            min_y = (y > 0) ? y : -y;
            root_x = x;
        }
    }

    std::cout << std::fixed << std::setprecision(5) << area << std::endl;
    return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.